

Occurrence Data

ADAE and the safety tables it feeds

One row per event. One flag decides what every safety table in the study reports.

By the end of this module you can...

1

Recognise an occurrence dataset and pick OCCDS vs BDS by SHAPE, not by subject matter.

2

Derive the analysis variables ASTDY, ADURN, ASEVN, AREL — and know why each exists.

3

Explain treatment-emergent and why ADAE reads the flag rather than recomputing it.

4

Set occurrence flags correctly so that counting flagged rows counts subjects.

5

Defend a causality definition and say who decides it and when.

Pick by shape, not by subject matter

Does it repeat over visits?

blood pressure • haemoglobin • a questionnaire score

BDS

Did it either occur, or not?

adverse event • concomitant medication • protocol deviation

OCCDS

Is it one fact about the subject?

treatment arm • age group • date of first dose

ADSL

A vital sign measured only once is still BDS. An adverse event is still OCCDS even though it has dates. The shape decides.

One row decides what every safety table reports

ABC-01-01-002 · AESEQ 1 · Oropharyngeal pain

Started study day -5. Five days BEFORE the first dose.

A real event, reported by a real investigator.

TRTEMFL = 'N'

It STAYS in the dataset

Deleting it breaks reconciliation with SDTM AE, and destroys every non-safety use of the data.

It appears in NO safety table

Every safety table subsets on TRTEMFL = 'Y'. The drug cannot have caused what preceded it.

It carries NO occurrence flag

AOCCFL and AOCCPFL are restricted to emergent events, or the flag would undo its own purpose.

Keep the row. Flag it. Subset at analysis time. That sentence is most of what ADaM is.

The flag already exists — read it, do not rebuild it

SDTM did this already

SUPPAE.AETRTEM

Derived once, in the tabulation package, where it is documented and where a reviewer can find it.

ADaM reads it

trtemfl = qval;

One derivation. One home.
If SDTM's rule is later refined,
ADaM inherits the refinement.

Recompute it in ADaM and the two WILL eventually disagree. Someone refines the SDTM rule for partial dates; ADaM keeps its own copy. Nothing errors. The tabulation and analysis packages simply disagree about which events were emergent — and a reviewer finds it before you do.

But trust is not verification. The notebook independently cross-checks the flag against the dates and requires zero mismatches. Reading a derived flag is right; assuming it is correct without ever testing it is not.

Why AOCCFL exists

The first line of every AE table is “subjects with at least one treatment-emergent event”.

A subject with three events must contribute 1 to that number — not 3.

Count DISTINCT subjects
where TRTEMFL = 'Y'

3 / 3

correct

Count ROWS
where AOCCFL = 'Y'

3 / 3

correct — by construction

Count ALL rows
where TRTEMFL = 'Y'

5 / 4

wrong — 125% of the arm

Drug A / Placebo. Four subjects per arm.

Count subjects with a flag. Count events with a row. Never let one stand in for the other. A percentage over 100 announces itself — the dangerous version is a study large enough that the inflated number still looks plausible.

Which row gets flagged must not depend on luck

FRAGILE

```
by usubjid astdt;
```

Ties unresolved.

The flagged row depends on
whatever order the data
happened to arrive in.

CORRECT

```
by usubjid astdt aseq;
```

Ties broken deterministically.
Same answer every run,
on every machine,
in every order.

Why this is a regulatory concern, not tidiness

1. **Reproducibility IS the deliverable.** A result that changes with dataset order cannot be reproduced — and an irreproducible result is treated as unreliable whether or not it was right.
2. **It breaks silently and intermittently.** Most damagingly between the run that made the tables and the run that made the QC check.
3. **The flagged row supplies reported values.** In some shells the flagged event's severity is what prints. A non-deterministic flag changes reported severities, not just row selection.

No ABC-01 subject has two events on one date — so this code would pass today and fail on the next study.

CAUSALITY

**AREL collapses four collected values into two.
That collapse is a decision, not a fact.**

RELATED or POSSIBLY RELATED

the conservative reading — and the usual default

7 of 9

RELATED only

22% of treatment-emergent events reclassified

5 of 9

Same events. Same investigator assessments. A different reading rule — and the headline rate moves.

Decided in the SAP, before database lock, before anyone has seen unblinded results.

WHAT'S NEXT

Ten rows, and one of them matters most

1

Step 1 • Notebook 15 Build ADAE. Merge ADSL, derive the analysis variables, set the occurrence flags.

2

Step 2 • Exercise 2 Delete the pre-dose event instead of flagging it. Watch nothing change — and understand why that is the problem.

3

Step 3 • Check PROC COMPARE against data/adam/adae.csv, and prove the flag arithmetic.

Two subjects have no events at all — and they still count in every denominator. That idea returns with force in ADLB.